

Untitled hardware project

Class auv	Feasibility feasible	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Survey endurance	8	hours	medium	Mission endurance target from the brief, before hydrodynamic drag, hotel-load and reserve-energy calculations.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class auv (high); requirements 2; numeric facts 2; requirement trace 2/2 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	209 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	2 quantified requirement(s), 2 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected auv with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	auv has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	2 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class auv (high); 2 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s): 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class auv; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 9; submodules 27; components 108.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 8/8.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class auv; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 240; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class auv; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	463 provenance ref(s); 240 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	108 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufactur er/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifi er lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 9/27/108. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.75	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/9 modules pass; 7 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 8/8 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 2/2 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 1; assumptions needing review 22.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	108 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 48 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_review_required

6/7 candidate areas pass; 1 review, 0 blocked. Candidate lines 108; critical 12/12; worklist rows 108; duplicate groups 0; provenance violations 0; ready for sourcing no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	108 candidate BoM line(s), 108 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	review	0.99	107/108 candidate line(s) have non-generic identity, description and role context.	sensing_instrumentation_navigation_sensor_suite_dvl: weak candidate identity for dvl.	Rename weak component candidates and add role context before sourcing them.
quantity_basis	pass	1	108/108 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	12/12 class-critical part(s) are represented by candidate lines; 12 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	108/108 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorization_review_required

5/6 sourcing-authorization areas pass; 1 review, 0 blocked. Critical intake 12/12; full intake 108/108; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	review	0.86	component_candidates_review_required ; 108 candidate line(s), 108 sourcing-worklist row(s), 0 provenance violation(s).	none	Resolve candidate review rows before admitting sourcing evidence.
critical_intake_scope	pass	1	12/12 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	108/108 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_review_required

Display mode candidate_only. 1/7 BoM-admission areas pass; 6 review, 0 blocked. Critical priced 0/12; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	review	0.83	sourcing_authorization_review_required ; authorized no; 5/6 areas pass.	none	Keep BoM admission review-only until sourcing authorization clears.
candidate_bom_scope	pass	1	108 candidate BoM line(s), 12 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/108 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/12 critical line(s) have admitted unit costs; 12 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.91	0 source-backed claim(s), 48 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 12. Verification accepted: 0/30; unaccepted: 30. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 12 critical BoM line(s) lack source-backed cost evidence.
- 7 verification activity/activities remain blocked.
- 0/30 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	12/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 9/27/108. Calculations within envelope 1/2; calculation blockers 0. Assumption review rows 22; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 27 submodules and 108 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	27/27 submodules ready; 12/12 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	8/8 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	2/2 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.5	1/2 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.31	10/32 assumptions are brief-supported or model-present; 22 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/12; unpriced critical lines 12. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/108 BoM lines have admitted pricing; rejected source records 0.	structure_containment_pressure_hull_s hell_pressure_hull: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_subsea_battery_pack_battery_pack : Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_dc_distribution_wet_mate_connector_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_thruster_array_thruster_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. control_compute_communication_mission_computer_stack_mission_computer: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 7 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/12 critical lines have admitted unit costs.	structure_containment_pressure_hull_s hell_pressure_hull: Pressure hull energy_storage_source_subsea_battery_pack_battery_pack : Battery pack power_distribution_dc_distribution_wet_mate_connector_set: Wet-mate connector set actuation_kinematics_thruster_array_thruster_set: Thruster set control_compute_communication_mission_computer_stack_mission_computer:	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				Mission computer plus 7 more	
source_backed_provenance	blocked	0.91	0 source-backed claims; 48 critical missing-source claims; 0 provenance violations.	structure_containment_pressure_hull_s hell_pressure_hull/supplier: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull_s hell_pressure_hull/ manufacturer: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull_s hell_pressure_hull/ mpn: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull_s hell_pressure_hull/u nit_cost_gbp: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_source_subsea_battery_pack_battery_pack /supplier: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 43 more	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	12 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

47 missing-evidence row(s): 15 sourcing, 32 verification. Ready 44; blocked 0. Blocks BoM 15; blocks publish 47. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	pressure_hull	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	battery_pack	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	wet_mate_c onnector_se t	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
9	sourcing	blocker	ready_for_in take	thruster_set	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
10	sourcing	blocker	ready_for_in take	mission_co mputer	supplier_cat alogue_or_q uote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_in take	acoustic_modem	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_in take	dvl	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_in take	inertial_navigation_unit	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_in take	depth_and_ pressure_se nsor	supplier_cat alogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
15	sourcing	blocker	ready_for_in take	forward_son ar_payload	supplier_cat alogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
16	sourcing	blocker	ready_for_intake	leak_detection_sensor	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
17	sourcing	blocker	ready_for_intake	recovery_beacon	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
18	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	calculation:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef;	anonymous reviewer; unchecked model assertion; test-fixture://	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidenceNote; reviewedAt	review; source-evidence claims that belong in sourcing intake	
20	verification	major	ready_for_intake	design_review:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	calculation:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	calculation:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
							sourcing intake	
24	verification	major	ready_for_intake	design_review:actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	calculation: actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
28	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	calculation:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	calculation:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
33	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	interface_review:energy_storage_source:power_distribution:dc_power_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
35	verification	major	ready_for_intake	interface_review:power_distribution:actuation_kinematics:thruster_power_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_intake	interface_review:control_compute_communication:actuation_kinematics:thrust_command_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
37	verification	major	ready_for_in take	interface_re view:sensin g_instrume ntation:cont rol_comput e_communi cation:navig ation_senso r_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
38	verification	major	ready_for_in take	interface_re view:sensin g_instrume ntation:cont rol_comput e_communi cation:paylo ad_data_bu s	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_in take	interface_re view:enviro nmental_int erface:struc ture_contai nment:press ure_bounda ry	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
40	verification	major	ready_for_intake	interface_review:safety_protection:control_compute_communication:alarm_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
41	verification	major	ready_for_intake	interface_review:safety_protection:structure_containment:pressure_boundary	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
42	verification	major	ready_for_intake	compliance_review:PRESSURE-BOUNDARY-REVIEW	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
43	verification	major	ready_for_intake	compliance_review:SUBSEA-LEAK-ABORT-REVIEW	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
44	verification	major	ready_for_intake	compliance_review:EMC-MARINE-	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
				COMMS- REVIEW		me; verdict; evidenceRef ; evidenceNo te; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
45	verification	major	ready_for_in take	risk_review: 1	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
46	verification	major	ready_for_in take	risk_review: 2	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
47	verification	major	ready_for_in take	risk_review: 3	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/30. Source-backed BoM claims 0; critical missing BoM claims 48; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s).

- engineering_math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom_sourcing has 60 blocked/source-required claim(s) and 96 review-required claim(s).
- compliance_risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).
- structure_containment_pressure_hull_shell_pressure_hull/supplier: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- structure_containment_pressure_hull_shell_pressure_hull/manufacture: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- structure_containment_pressure_hull_shell_pressure_hull/mpn: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- structure_containment_pressure_hull_shell_pressure_hull/unit_cost_gbp: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_subsea_battery_pack_battery_pack/supplier: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_subsea_battery_pack_battery_pack/manufacture: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_subsea_battery_pack_battery_pack/mpn: Admit source-backed supplier/manufacture/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 27 submodules and 8 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	2/205 claims pass; 62 blocked/source-required, 141 review-required.	architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 60 blocked/source-required claim(s) and 96 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/2 requirements have accepted assurance; 2 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/30 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 30 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.91	0 source-backed BoM claims, 48 critical missing-source claims, 0 provenance violations.	structure_containment_pressure_hull/shell_pressure_hull/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull/shell_pressure_hull/manufacturer:	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull_shell_pressure_hull/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_pressure_hull_shell_pressure_hull/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_subsea_battery_pack_battery_pack/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 43 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 57; trust blockers 61. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	15	52	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	9	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	32	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

205 claims: 2 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 138 generated needing review, 62 source-required, 0 blocked. Blocking: 62 BoM, 202 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Thruster set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thruster set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thruster set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thruster set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Acoustic modem / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Acoustic modem / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Acoustic modem / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Acoustic modem / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Mission computer / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Mission computer / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Mission computer / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Mission computer / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Battery pack / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Battery pack / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Battery pack / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Battery pack / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Wet-mate connector set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Wet-mate connector set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Wet-mate connector set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Wet-mate connector set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery beacon / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery beacon / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery beacon / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery beacon /	unit_cost_gbp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			unit_cost_gbp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Leak detection sensor / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Leak detection sensor / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Leak detection sensor / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Leak detection sensor / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Depth and pressure sensor / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Depth and pressure sensor / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Depth and pressure sensor / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Depth and pressure sensor / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DVL / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DVL / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DVL / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	DVL / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inertial navigation unit / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inertial navigation unit / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inertial navigation unit / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inertial navigation unit / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Forward sonar payload / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Forward sonar payload / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Forward sonar payload / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Forward sonar payload / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Pressure hull / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Pressure hull / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Pressure hull / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Pressure hull / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit- cost evidence before BoM can pass.
component_candidate	source_required	generated_model	Thruster set	Thruster set is an architecture-derived component candidate with quantity 6 each.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Acoustic modem	Acoustic modem is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Mission computer	Mission computer is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Battery pack	Battery pack is an architecture-derived	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier,

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Wet-mate connector set	component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Wet-mate connector set is an architecture-derived component candidate with quantity 1 candidate.				Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Recovery beacon is an architecture-derived component candidate with quantity 1 candidate.				Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Leak detection sensor is an architecture-derived component candidate with quantity 1 candidate.				Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Depth and pressure sensor	Depth and pressure sensor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	DVL	DVL is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Inertial navigation unit	Inertial navigation unit is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Forward sonar payload	Forward sonar payload is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Pressure hull	Pressure hull is an architecture-derived component candidate with	model:scratch_universal_architecture.auv	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:auv.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	controller heat spreader	controller heat spreader is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	current sensor channel set	current sensor channel set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	thrust calibration table	thrust calibration table is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	thruster motor controller stack	thruster motor controller stack is an architecture-derived component	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				candidate with quantity 1 candidate.				installation instance.
component_candidate	generated_needs_review	generated_model	motor phase harness	motor phase harness is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	thruster mounting bracket	thruster mounting bracket is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	thruster nozzle guard	thruster nozzle guard is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	adjustable trim fin	adjustable trim fin is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.auv	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 205 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

2/205 claims pass by being brief-supplied, source-backed or reviewer-accepted. 62 blocked/source-required claims, 141 review-required claims. Claim pass ratio 0.01. Blocking: 62 BoM, 202 publishable.

- architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 60 blocked/source-required claim(s) and 96 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	2	2	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	36	0	36	0	0	0	36	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	4	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	156	0	96	60	60	60	156	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	6	0	6	0	0	0	6	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

57 open evidence gaps: 18 blockers, 39 major, 0 minor. Closure paths: 15 sourcing intake, 32 verification intake, 9 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 15, publishable 57.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	12 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	7 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/30 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	AUV endurance minutes	480 minutes mission endurance before energy-budget proof.	engineering_review	publishable	Confirm speed profile, payload load, reserve margin, battery derating and recovery scenario.
assumption_review	major	needs review	Survey endurance	Survey endurance uses endurance_hours with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Depth Rating	Depth Rating currently evaluates as pass against 10 to 600 m for inspection-class AUV prototypes; deeper missions need dedicated pressure-vessel, seal and recovery evidence.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Survey Endurance Target	Survey Endurance Target currently evaluates as pass against	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				<=12 h plausible for compact inspection AUV concept, 12-24 h needs battery/drag budget, >24 h is high risk without a detailed energy model.			backed rationale for this envelope.
assumption_review	major	needs review	Navigation/Control Closure	Navigation/Control Closure currently evaluates as pass against navigation sensor bus, payload data bus, acoustic link and thrust command bus present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Pressure/Power/Recovery Safety	Pressure/Power/Recovery Safety currently evaluates as pass against pressure boundary, leak alarm, alarm bus, DC power, service access and recovery beacon present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	PRESSURE-BOUNDARY-REVIEW	Depth-rated pressure boundary review is relevant to this product class in Project engineering gate.	engineering_review	publishable	Depth rating calculation, hull material allowables, endcap seal design and proof-pressure plan.
assumption_review	major	needs review	SUBSEA-LEAK-ABORT-REVIEW	Leak detection and abort-state review is relevant to this product class in Project engineering gate.	engineering_review	publishable	Leak sensor placement, abort state machine, battery isolation and recovery procedure.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
assumption_review	major	needs review	EMC-MARINE-COMMS-REVIEW	Marine communications and EMC review is relevant to this product class in Project engineering gate.	engineering_review	publishable	Acoustic modem integration, antenna/transducer mounting, cable shielding and data-loss behaviour.
sourcing_evidence	blocker	ready for intake	Pressure hull	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Battery pack	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Wet-mate connector set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Thruster set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Mission computer	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Acoustic modem	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	DVL	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Inertial navigation unit	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Depth and pressure sensor	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Forward sonar payload	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Leak detection sensor	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Recovery beacon	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Pressure Structure Containment	Review Pressure Structure Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Pressure Structure Containment	Verify Pressure Structure Containment	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				requirement calculations			verification intake template.
verification_evidence	major	ready for intake	Subsea Energy Storage	Review Subsea Energy Storage architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Subsea Energy Storage	Verify Subsea Energy Storage requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Subsea Power Distribution	Review Subsea Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Subsea Power Distribution	Verify Subsea Power Distribution requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thruster Actuation Kinematics	Review Thruster Actuation Kinematics architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thruster Actuation Kinematics	Verify Thruster Actuation Kinematics requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Autonomy Compute Communication	Review Autonomy Compute Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Autonomy Compute Communication	Verify Autonomy Compute Communication	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				n requirement calculations			verification intake template.
verification_evidence	major	ready for intake	Navigation And Payload Sensing	Review Navigation And Payload Sensing architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Pressure Thermal Buoyancy Interface	Review Pressure Thermal Buoyancy Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Pressure Thermal Buoyancy Interface	Verify Pressure Thermal Buoyancy Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Leak Abort And Recovery Safety	Review Leak Abort And Recovery Safety architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Leak Abort And Recovery Safety	Verify Leak Abort And Recovery Safety requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Deck Recovery And Service	Review Deck Recovery And Service architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Subsea Energy Storage -> Subsea Power Distribution	Verify dc_power_buss interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Subsea Power Distribution -> Thruster	Verify thruster_power_bus	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
			Actuation Kinematics	interface contract			verification intake template.
verification_evidence	major	ready for intake	Autonomy Compute Communication -> Thruster Actuation Kinematics	Verify thrust_command_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Navigation And Payload Sensing -> Autonomy Compute Communication	Verify navigation_sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Navigation And Payload Sensing -> Autonomy Compute Communication	Verify payload_data_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Pressure Thermal Buoyancy Interface -> Pressure Structure Containment	Verify pressure_boundary interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Leak Abort And Recovery Safety -> Autonomy Compute Communication	Verify alarm_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Leak Abort And Recovery Safety -> Pressure Structure Containment	Verify pressure_boundary interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for PRESSURE-BOUNDARY-REVIEW	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for SUBSEA-LEAK-ABORT-REVIEW	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for EMC-MARINE-COMMS-REVIEW	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Pressure hull or seal failure at depth	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Loss of navigation or communications underwater	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Unable to recover vehicle after abort	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 15, engineering 9, verification 32, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	15	15/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:acoustic_modem; sourcing:battery_pack; sourcing:depth_and_pressure_sensor; sourcing:dvl; sourcing:forward_sonar_payload; plus 7 more
3	Complete Engineering Review	9	0/9/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:EMC-MARINE-COMMS-REVIEW; assumption:compliance:PRESSURE-BOUNDARY-REVIEW; assumption:compliance:SUBSEA-LEAK-ABORT-REVIEW; assumption:metric:headline_output; assumption:safety:auv_depth_rating; assumption:safety:auv_endurance_target; assumption:safety:auv_navigation_control_closure; assumption:safety:auv_pressure_recovery_safety; plus 1 more
4	Accept Reviewer Evidence	32	2/30/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation_kinematics; verification:calculation_control_communication;

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							verification:calculation:energy_storage_source; verification:calculation:environmental_interface; verification:calculation:power_distribution; verification:calculation:safety_protection; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 0.96.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	9 modules	9 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	27 submodules	27 submodules	1	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	108 components	108 components	1	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	42 links	34 links	1.24	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	37 activities	22 activities	1.68	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a 300 m depth-rated autonomous underwater vehicle with 8 hour survey endurance, pressure hull, thruster set, DVL, battery pack, acoustic modem, leak detection, forward sonar payload and recovery beacon.

Requirement	Value	Unit
Endurance	8	hours
Depth rating	300	m

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Endurance	covered	8 hours	Subsea Energy Storage / Subsea battery pack; Subsea Power Distribution / DC distribution; Thruster Actuation Kinematics / Thruster array; Autonomy Compute Communication / Mission computer stack	metric:headline_out put; sanity:auv_endurance_target; sanity:auv_navigation_control_closure; sanity:auv_pressure_power_recovery_safety	Endurance is traced through battery, power distribution, propulsion and the autonomy logic that preserves reserve energy. Requirement has architecture coverage plus metric or engineering sanity coverage.
Depth rating	covered	300 m	Pressure Structure Containment / Pressure hull shell; Pressure Thermal Buoyancy Interface / Pressure boundary management; Leak Abort And Recovery Safety / Leak detection	sanity:auv_depth_rating; sanity:auv_pressure_power_recovery_safety	Depth rating is traced through the pressure hull, boundary support hardware and pressure-related safety checks. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

2 requirement row(s): 0 accepted with evidence, 2 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Depth rating	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; environmental_interface; safety_protection	design_review:structure_containment; calculation:structure_containment; design_review:environmental_interface; calculation:environmental_interface; design_review:safety_protection; calculation:safety_protection	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Endurance	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_storage_source; power_distribution; actuation_kinematics; control_compute_communication	design_review:energy_storage_source; calculation:energy_storage_source; design_review:power_distribution; calculation:power_distribution; design_review:actuation_kinematics; calculation:actuation_kinematics; design_review:control_compute_communication; calculation:control_compute_communication	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
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Engineering Assurance Matrix

2 requirements: 0 accepted, 0 ready, 2 needing review, 0 blocked, 0 unlinked. 2 have calculation links, 2 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Depth rating	needs review	covered	Pressure Structure Containment; Pressure Thermal Buoyancy Interface; Leak Abort And Recovery Safety	auv_hydrostatic_pressure_bar	assumption_resolution: component: safety_protection_abort_recovery_behavior; assumption_resolution: component: safety_protection_leak_detection_leak_detection_sensor; assumption_resolution: component: structure_containment_pressure_hull_shell_pressure_hull; assumption_resolution: sanity:auv_depth_rating; assumption_resolution: sanity:auv_pressure_power_recovery_safety; plus 16 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
Endurance	needs review	covered	Subsea Energy Storage; Subsea Power Distribution; Thruster Actuation Kinematics; Autonomy Compute Communication	auv_endurance_minutes	assumption_resolution: component: actuation_kinematics_thruster_array_thruster_set; assumption_resolution: component: control_communication_acoustic_communications_acoustic_modem; assumption_resolution: component: control_communication_mission_computer_stack_mission_co	pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
					computer; assumption _resolution: component: energy_storage_source _subsea_battery_pack_battery_pack; assumption _resolution: component: power_distribution_dc_distribution_wet_mate_connector_set; plus 26 more			

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 2; numeric_fact_count : 2; requirement_trace_covered: 2; requirement_trace_partial: 0; requirement_trace_uncovered: 0	endurance_hours: 8 hours; depth_rating_m: 300 m; endurance_hours_covered; depth_rating_m_covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected auv with high confidence from brief keywords.	selected_class: auv; confidence: high; keyword_scores: auv:6	auv: 6	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 9-module auv architecture from the scratch universal architecture grammar.	module_count: 9; architecture_source : scratch_universal_architecture	Pressure Structure Containment; Subsea Energy Storage; Subsea Power Distribution; Thruster Actuation Kinematics; Autonomy Compute Communication; Navigation And Payload Sensing; plus 3 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 27; average_submodules_per_module: 3	structure_containment.pressure_hull_shell; structure_containment.internal_mounting_frame; structure_containment.hydrodynamic_fairing; energy_storage_source.subsea_battery_pack; energy_storage_source.charging_interface; energy_storage_source.energy_monitoring; plus 21 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 8; present_links: 8; missing_links: 0; graph_nodes: 36; graph_edges: 69; shared_interface_edges: 34; missing_required_graph_edges: 0	energy_storage_source -> power_distribution via dc_power_bus: present; power_distribution -> actuation_kinematics via thruster_power_bus: present; control_compute_c	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				ommunication -> actuation_kinematic s via thrust_command_b us: present; sensing_instrument ation -> control_compute_c ommunication via navigation_sensor_ bus: present; sensing_instrument ation -> control_compute_c ommunication via payload_data_bus: present; environmental_inter face -> structure_containm ent via pressure_boundary: present; plus 2 more	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 108; critical_candidate_li nes: 12; admitted_price_line s: 0	pressure_hull: Pressure hull; endcap_clamp_ring: endcap clamp ring; double_o_ring_glan d_set: double O- ring gland set; hull_proof_test_por t: hull proof-test port; electronics_tray_rail : electronics tray rail; battery_cradle_fram e: battery cradle frame; plus 24 more	Showing first 30 of 108 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_cou nt: 0; sanity_pass: 4; sanity_warn: 0; sanity_fail: 0	pass Depth Rating: 300 m; pass Survey Endurance Target: 8 hours; pass Navigation/Control Closure: present; pass Pressure/Power/Rec overy Safety: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign- off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 108; admitted_lines: 0; unpriced_lines: 108; unpriced_critical_lin es: 12; rejected_source_re cords: 0	none	BoM cost remains intentionally blocked until 12 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

9 modules, 27 sub-modules, 108 allocated component words.

From	To	Interface	Status	Engineering Reason
energy_storage_source	power_distribution	dc_power_bus	OK	Battery energy must enter protected DC distribution through an explicit bus.
power_distribution	actuation_kinematics	thruster_power_bus	OK	Thrusters must receive protected power separately from avionics and payload rails.
control_compute_communication	actuation_kinematics	thrust_command_bus	OK	Mission control must command thruster drive states over a declared control interface.
sensing_instrumentation	control_compute_communication	navigation_sensor_bus	OK	DVL, INS and depth data must feed navigation fusion before mission autonomy is credible.
sensing_instrumentation	control_compute_communication	payload_data_bus	OK	Inspection payload data must be logged or acted on by the mission computer.
environmental_interface	structure_containment	pressure_boundary	OK	Hull, penetrators, thermal path and buoyancy hardware must share one depth-rated boundary model.
safety_protection	control_compute_communication	alarm_bus	OK	Leak, over-depth and recovery alarms must drive fail-safe mission states.
safety_protection	structure_containment	pressure_boundary	OK	Leak and over-depth protection must reference the same pressure boundary as the hull design.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
9	27	34	8	0

Interface Contract Matrix

8/8 required contracts present. 13 shared interface families, 4 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
dc_power_bu s	Subsea Energy Storage	Subsea battery pack; Charging interface	Subsea Power Distribution	DC distribution; Isolation and grounding	present	Battery energy must enter protected DC distribution through an explicit bus.	Both endpoints expose the required interface through at least one submodule carrier.
thruster_powe r_bus	Subsea Power Distribution	DC distribution	Thruster Actuation Kinematics	Thruster array; Drive electronics	present	Thrusters must receive protected power separately from avionics and payload rails.	Both endpoints expose the required interface through at least one submodule carrier.
thrust_comma nd_bus	Autonomy Compute Communication	Mission computer stack	Thruster Actuation Kinematics	Thruster array; Drive electronics	present	Mission control must command thruster drive states over a declared control interface.	Both endpoints expose the required interface through at least one submodule carrier.
navigation_se nsor_bus	Navigation And Payload Sensing	Navigation sensor suite; Depth and environment sensors; Payload sensor bay	Autonomy Compute Communication	Mission computer stack	present	DVL, INS and depth data must feed navigation fusion before mission autonomy is credible.	Both endpoints expose the required interface through at least one submodule carrier.
payload_data _bus	Navigation And Payload Sensing	Payload sensor bay	Autonomy Compute Communication	Mission computer stack	present	Inspection payload data must be logged or acted on by the mission computer.	Both endpoints expose the required interface through at least one submodule carrier.
pressure_bou ndary	Pressure Thermal Buoyancy Interface	Pressure boundary management; Thermal path	Pressure Structure Containment	Pressure hull shell	present	Hull, penetrators, thermal path and buoyancy hardware must share one depth- rated boundary model.	Both endpoints expose the required interface through at least one submodule carrier.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
alarm_bus	Leak Abort And Recovery Safety	Leak detection; Abort and recovery; Pressure fail- safe	Autonomy Compute Communicatio n	Mission computer stack; Acoustic communicatio ns; Software health management	present	Leak, over- depth and recovery alarms must drive fail-safe mission states.	Both endpoints expose the required interface through at least one submodule carrier.
pressure_bou ndary	Leak Abort And Recovery Safety	Leak detection; Pressure fail- safe	Pressure Structure Containment	Pressure hull shell	present	Leak and over-depth protection must reference the same pressure boundary as the hull design.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: buoyancy_trim (Pressure Thermal Buoyancy Interface); leak_alarm_bus (Leak Abort And Recovery Safety); low_voltage_rail (Subsea Power Distribution); payload_power_bus (Subsea Power Distribution).

Interface Verification Gate

interface_review_ready

8 required interface row(s): 0 accepted, 8 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
dc_power_buses	Subsea Energy Storage	Subsea Power Distribution	both_endpoints_carried	interface_review:energy_storage_source:power_distribution:dc_power_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
thruster_power_bus	Subsea Power Distribution	Thruster Actuation Kinematics	both_endpoints_carried	interface_review:power_distribution:actuation_kinematics:thruster_power_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
thrust_command_bus	Autonomy Compute Communication	Thruster Actuation Kinematics	both_endpoints_carried	interface_review:control_compute_communication:actuation_kinematics:thrust_command_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
navigation_sensor_bus	Navigation And Payload Sensing	Autonomy Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:navigation_sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
payload_data_bus	Navigation And Payload Sensing	Autonomy Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:payload_data_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
pressure_boundary	Pressure Thermal Buoyancy Interface	Pressure Structure Containment	both_endpoints_carried	interface_review:environmental_interface:structure_containment:pressure_boundary	pending	review_ready	Collect named interface-review evidence through verification intake.
alarm_bus	Leak Abort And Recovery Safety	Autonomy Compute Communication	both_endpoints_carried	interface_review:safety_protection:control_compute_communication:alarm_bus	pending	review_ready	Collect named interface-review evidence through verification intake.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
pressure_boundary	Leak Abort And Recovery Safety	Pressure Structure Containment	both_endpoints_carried	interface_review:safety_protection:structure_containment:pressure_boundary	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

27/27 submodules ready; 0 review, 0 blocked. Component words 108; critical parts 12/12; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Pressure Structure Containment	Pressure hull shell	ready	4	pressure_boundary; mechanical_mounts; service_access	pressure_hull	none	Submodule has component candidates and no allocation blocker.
Pressure Structure Containment	Internal mounting frame	ready	4	mechanical_mounts; service_access	none	none	Submodule has component candidates and no allocation blocker.
Pressure Structure Containment	Hydrodynamic fairing	ready	4	hydrodynamic_shell; mechanical_mounts; service_access	none	none	Submodule has component candidates and no allocation blocker.
Subsea Energy Storage	Subsea battery pack	ready	4	dc_power_buses; battery_monitor_bus; thermal_path	battery_pack	none	Submodule has component candidates and no allocation blocker.
Subsea Energy Storage	Charging interface	ready	4	service_access; dc_power_buses; battery_monitor_bus	none	none	Submodule has component candidates and no allocation blocker.
Subsea Energy Storage	Energy monitoring	ready	4	battery_monitor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Subsea Power Distribution	DC distribution	ready	4	dc_power_buses; thruster_power_bus; payload_power_bus	wet_mate_connector_set	none	Submodule has component candidates and no allocation blocker.
Subsea Power Distribution	Low-voltage rails	ready	4	low_voltage_rail; payload_power_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Subsea Power Distribution	Isolation and grounding	ready	4	dc_power_bu s; alarm_bus; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Thruster Actuation Kinematics	Thruster array	ready	4	thruster_powe r_bus; thrust_comma nd_bus; mechanical_m ounts	thruster_set	none	Submodule has component candidates and no allocation blocker.
Thruster Actuation Kinematics	Drive electronics	ready	4	thruster_powe r_bus; thrust_comma nd_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Thruster Actuation Kinematics	Trim and stability surfaces	ready	4	hydrodynamic _shell; mechanical_m ounts; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Autonomy Compute Communication	Mission computer stack	ready	4	navigation_se nsor_bus; payload_data _bus; thrust_comma nd_bus; alarm_bus	mission_comp uter	none	Submodule has component candidates and no allocation blocker.
Autonomy Compute Communication	Acoustic communications	ready	4	acoustic_link; service_acces s; alarm_bus	acoustic_mod em	none	Submodule has component candidates and no allocation blocker.
Autonomy Compute Communication	Software health management	ready	4	alarm_bus; battery_monit or_bus; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Navigation And Payload Sensing	Navigation sensor suite	ready	4	navigation_se nsor_bus; service_acces s	dvl; inertial_navig ation_unit	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Navigation And Payload Sensing	Depth and environment sensors	ready	4	navigation_sensor_bus; pressure_boundary; alarm_bus	depth_and_pressure_sensor	none	Submodule has component candidates and no allocation blocker.
Navigation And Payload Sensing	Payload sensor bay	ready	4	payload_data_bus; navigation_sensor_bus; pressure_boundary	forward_sonar_payload	none	Submodule has component candidates and no allocation blocker.
Pressure Thermal Buoyancy Interface	Pressure boundary management	ready	4	pressure_boundary; service_access	none	none	Submodule has component candidates and no allocation blocker.
Pressure Thermal Buoyancy Interface	Thermal path	ready	4	thermal_path; pressure_boundary; mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Pressure Thermal Buoyancy Interface	Buoyancy and trim	ready	4	buoyancy_trim; hydrodynamic_shell; service_access	none	none	Submodule has component candidates and no allocation blocker.
Leak Abort And Recovery Safety	Leak detection	ready	4	leak_alarm_bus; alarm_bus; pressure_boundary	leak_detection_sensor	none	Submodule has component candidates and no allocation blocker.
Leak Abort And Recovery Safety	Abort and recovery	ready	4	alarm_bus; service_access; acoustic_link	recovery beacon	none	Submodule has component candidates and no allocation blocker.
Leak Abort And Recovery Safety	Pressure fail-safe	ready	4	pressure_boundary; dc_power_buses; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Deck Recovery And Service	Deck recovery	ready	4	mechanical_m ounts; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Deck Recovery And Service	Service access	ready	4	service_acces s; pressure_bou ndary; dc_power_bu s	none	none	Submodule has component candidates and no allocation blocker.
Deck Recovery And Service	Field transport	ready	4	service_acces s; hydrodynamic _shell	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/27 submodules pass; 27 review, 0 blocked. Component words 108; linked requirements 7; carrier contracts 28/28; local-only interfaces 49. Review questions accepted 0/82; verification accepted 0/73; critical unpriced lines 12.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Pressure Structure Containment	Pressure hull shell	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Pressure Structure Containment	Internal mounting frame	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 2 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 1 verification activity row(s) await accepted evidence.
Pressure Structure Containment	Hydrodynamic fairing	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): hydrodynamic_s hell, mechanical_mounts, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Subsea Energy Storage	Subsea battery pack	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): battery_monitor_bus, thermal_path. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Subsea Energy Storage	Charging interface	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): service_access, battery_monitor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Subsea Energy Storage	Energy monitoring	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): battery_monitor

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											_bus, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Subsea Power Distribution	DC distribution	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): payload_power_bus. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Subsea Power Distribution	Low-voltage rails	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): low_voltage_rail, payload_power_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Subsea Power Distribution	Isolation and grounding	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): alarm_bus, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Thrustor Actuation Kinematics	Thrustor array	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): mechanical_mounts. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Thrustor Actuation Kinematics	Drive electronics	review	4	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface declaration(s): alarm_buss. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await
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Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											accepted evidence.
Thrustor Actuation Kinematics	Trim and stability surfaces	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): hydrodynamic_s hell, mechanical_mounts, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance evidence.
Autonomy Computer Communication	Mission computer stack	review	4	4	1	4/4	0/6	0/7	1	0.6667	6 engineering review question(s) await acceptance. 6 verification activity row(s) await acceptance evidence. 1 critical component line(s) still need source-

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											backed cost/manufacturer/MPN evidence.
Autonomous Compute Communication	Acoustic communications	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): acoustic_link, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Autonomous Compute Communication	Software health management	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): battery_monitor_bus, service_access. 3 engineering

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ring review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Navigation And Payload Sensing	Navigation sensor suite	review	4	2	0	1/1	0/3	0/3	2	0.6667	Review 1 local-only interface declaration(s): service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Navigation And Payload Sensing	Depth and environment sensors	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion(s): pressure_boundary, alarm_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Navigation And Payload Sensing	Payload sensor bay	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): pressure_boundary. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Pressure Thermal Buoyancy Interface	Pressure boundary management	review	4	2	1	1/1	0/3	0/3	0	0.6667	Review 1 local-only interface declaration(s): service_3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Pressure Thermal Buoyancy Interface	Thermal path	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): thermal_path, mechanical_mounts. 3 engineering review question(s) await acceptance. 2 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence.
Pressure Thermal Buoyancy Interface	Buoyancy and trim	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): buoyancy_trim, hydrodynamic_s hell, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Leak Abort And Recovery Safety	Leak detection	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): leak_alarm_bus. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Leak Abort And Recovery Safety	Abort and recovery	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interfaces declaration(s): service_access, acoustic_link. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Leak Abort And	Pressure fail-safe	review	4	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Recovery Safety											engineering declaration(s): dc_power_bus. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Deck Recovery And Service	Deck recovery	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Deck Recovery And Service	Service access	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access,

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											pressure_boundary, dc_power_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Deck Recovery And Service	Field transport	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, hydrodynamic_s hell. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module Engineering Gate

module_engineering_review_required

0/9 modules pass; 9 review, 0 blocked. Submodules/components: 27/108. Interface carriers 16/16; review questions accepted 0/89; critical unpriced lines 12 across 7 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Pressure Structure Containme nt	review	3	12	1	2/2	0/10	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or source evidence.
Subsea Energy Storage	review	3	12	1	1/1	0/10	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or source evidence.
Subsea Power Distributio n	review	3	12	1	2/2	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Thruster Actuation Kinematic s	review	3	12	1	2/2	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Autonomy Compute Communic ation	review	3	12	1	4/4	0/14	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 9 assumptio n row(s) need review or source evidence.
Navigation And Payload Sensing	review	3	12	0	2/2	0/10	4	0.71	4 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									review or source evidence.
Pressure Thermal Buoyancy Interface	review	3	12	1	1/1	0/8	0	0.86	8 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Leak Abort And Recovery Safety	review	3	12	1	2/2	0/11	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or source evidence.
Deck Recovery And Service	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or source evidence.

Module Review Roll-Up

2 ready, 7 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Pressure Structure Containment	sourcing	3	12	2/2	depth_rating_m	pressure_hull	1 critical line(s) need source-backed evidence before BoM pass.
Subsea Energy Storage	sourcing	3	12	1/1	endurance_hours	battery_pack	1 critical line(s) need source-backed evidence before BoM pass.
Subsea Power Distribution	sourcing	3	12	2/2	endurance_hours	wet_mate_connector_set	1 critical line(s) need source-backed evidence before BoM pass.
Thruster Actuation Kinematics	sourcing	3	12	2/2	endurance_hours	thruster_set	1 critical line(s) need source-backed evidence before BoM pass.
Autonomy Compute Communication	sourcing	3	12	4/4	endurance_hours	mission_computer; acoustic_modem	2 critical line(s) need source-backed evidence before BoM pass.
Navigation And Payload Sensing	sourcing	3	12	2/2	none	dvl; inertial_navigation_unit; depth_and_pressure_sensor; forward_sonar_payload	4 critical line(s) need source-backed evidence before BoM pass.
Pressure Thermal Buoyancy Interface	ready	3	12	1/1	depth_rating_m	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Leak Abort And Recovery Safety	sourcing	3	12	2/2	depth_rating_m	leak_detection_sensor; recovery_beacon	2 critical line(s) need source-backed evidence before BoM pass.
Deck Recovery And Service	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

76 reviewer questions: 0 accepted, 45 ready, 31 needing review, 0 blocked. Coverage: 9 modules, 27 submodules, 8 interfaces, 2 calculations, 30 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumptio n_resoluti on	major	needs review	EMC-MARINE-COMMS-REVIEW	Can the EMC-MARINE-COMMS-REVIEW assumption be accepted for architecture review, or must it be replaced with evidence?	Acoustic modem integration , antenna/transducer mounting, cable shielding and data-loss behaviour.	none	none	none	none
assumptio n_resoluti on	major	needs review	PRESSUR E-BOUNDAR Y-REVIEW	Can the PRESSUR E-BOUNDAR Y-REVIEW assumption be accepted for architecture review, or must it be replaced with evidence?	Depth rating calculation , hull material allowables , endcap seal design and proof-pressure plan.	none	none	none	none
assumptio n_resoluti on	major	needs review	SUBSEA-LEAK-ABORT-REVIEW	Can the SUBSEA-LEAK-ABORT-REVIEW assumption be accepted for architecture review, or must it be replaced with evidence?	Leak sensor placement , abort state machine, battery isolation and recovery procedure.	none	none	none	none
assumptio n_resoluti on	major	needs review	Thruster set	Can the Thruster set assumption be accepted for architecture review, or must it be replaced	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through	actuation_ kinematics	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?	sourcing intake.				
assumption_resolution	major	needs review	Acoustic modem	Can the Acoustic modem assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Mission computer	Can the Mission computer assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Battery pack	Can the Battery pack assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_storage_source	none	none	none
assumption_resolution	major	needs review	Wet-mate connector set	Can the Wet-mate connector set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	power_distribution	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Recovery beacon	Can the Recovery beacon assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	safety_protection	none	none	none
assumption_resolution	major	needs review	Leak detection sensor	Can the Leak detection sensor assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	safety_protection	none	none	none
assumption_resolution	major	needs review	Depth and pressure sensor	Can the Depth and pressure sensor assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	DVL	Can the DVL assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Inertial navigation unit	Can the Inertial navigation unit assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Forward sonar payload	Can the Forward sonar payload assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Pressure hull	Can the Pressure hull assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> actuation_kinematics	Can the control_compute_communication -> actuation_kinematics assumption be accepted for architecture review, or must it be replaced	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thrust_command_buss	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	energy_storage_source -> power_distribution	Can the energy_storage_source -> power_distribution assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	dc_power_bus	none	none
assumption_resolution	major	needs review	environmental_interface -> structure_containment	Can the environmental_interface -> structure_containment assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	pressure_boundary	none	none
assumption_resolution	major	needs review	power_distribution -> actuation_kinematics	Can the power_distribution -> actuation_kinematics assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thruster_power_bus	none	none
assumption_resolution	major	needs review	safety_protection -> control_compute_communication	Can the safety_protection -> control_compute_communication	Review the interface carrier, direction, capacity, safety	none	alarm_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				assumption be accepted for architecture review, or must it be replaced with evidence?	margin and failure mode before detailed design or sourcing.				
assumption_resolution	major	needs review	safety_protection -> structure_containment	Can the safety_protection -> structure_containment assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	pressure_boundary	none	none
assumption_resolution	major	needs review	sensing_instrumentation -> control_compute_communication	Can the sensing_instrumentation -> control_compute_communication assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	navigation_sensor_buses	none	none
assumption_resolution	major	needs review	sensing_instrumentation -> control_compute_communication	Can the sensing_instrumentation -> control_compute_communication assumption be accepted for architecture review, or must it	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	payload_data_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				be replaced with evidence?					
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Survey endurance	Can the Survey endurance assumption be accepted for architecture review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	needs review	Depth Rating	Can the Depth Rating assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_containment; environmental_interface; safety_protection	none	none	none
assumption_resolution	major	needs review	Survey Endurance Target	Can the Survey Endurance Target assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_storage_source; power_distribution; actuation_kinematics ; control_compute_communication	none	none	none
assumption_resolution	major	needs review	Navigation /Control Closure	Can the Navigation /Control Closure assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_storage_source; power_distribution; actuation_kinematics ; control_compute_communication	none	none	none
assumption_resolution	major	needs review	Pressure/Power/Recovery Safety	Can the Pressure/Power/Recovery Safety assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_storage_source; power_distribution; actuation_kinematics ; control_compute_communication; structure_containment; environmental_interface	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
						ace; safety_protection			
calculation_envelope	major	needs review	AUV endurance minutes	Does the AUV endurance minutes calculation support the architecture envelope, or does the concept need revision?	Confirm speed profile, payload load, reserve margin, battery derating and recovery scenario.	energy_storage_source; power_distribution; actuation_kinematics; control_compute_communication	none	auv_endurance_minutes	none
calculation_envelope	major	ready	AUV hydrostatic pressure	Does the AUV hydrostatic pressure calculation support the architecture envelope, or does the concept need revision?	Confirm hull material, safety factor, penetrators, seals, buckling margin and test pressure.	structure_containment; environmental_interface; safety_protection	none	auv_hydrostatic_pressure_bar	none
interface_contract	major	ready	Autonomy Compute Communication -> Thruster Actuation Kinematics via thrust_command_buses	Is the thrust_command_buses interface between Autonomy Compute Communication and Thruster Actuation Kinematics correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; actuation_kinematics	thrust_command_buses	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
interface_contract	major	ready	Subsea Energy Storage -> Subsea Power Distribution via dc_power_bus	Is the dc_power_bus interface between Subsea Energy Storage and Subsea Power Distribution correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; power_distribution	dc_power_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Pressure Thermal Buoyancy Interface -> Pressure Structure Containment via pressure_boundary	Is the pressure_boundary interface between Pressure Thermal Buoyancy Interface and Pressure Structure Containment correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; structure_containment	pressure_boundary	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Subsea Power Distribution -> Thruster Actuation Kinematics via thruster_power_bus	Is the thruster_power_bus interface between Subsea Power Distribution and Thruster Actuation Kinematics correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	power_distribution; actuation_kinematics	thruster_power_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Leak Abort And Recovery Safety -> Autonomy Compute Communication	Is the alarm_bus interface between Leak Abort And Recovery Safety and	Interface-review evidence naming both endpoint owners, carrier	safety_protection; control_compute_communication	alarm_bus	none	Both endpoints expose the required interface through at least one

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
			ation via alarm_bus	Autonomy Compute Communication correctly owned and physically carried?	submodules, capacity limits and any deferred checks.				submodule carrier.
interface_contract	major	ready	Leak Abort And Recovery Safety -> Pressure Structure Containment via pressure_boundary	Is the pressure_boundary interface between Leak Abort And Recovery Safety and Pressure Structure Containment correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	safety_protection; structure_containment	pressure_boundary	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Navigation And Payload Sensing -> Autonomy Compute Communication via navigation_sensor_bus	Is the navigation_sensor_bus interface between Navigation And Payload Sensing and Autonomy Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_communication	navigation_sensor_bus	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
interface_contract	major	ready	Navigation And Payload Sensing - > Autonomy Compute Communication via payload_data_bus	Is the payload_data_bus interface between Navigation And Payload Sensing and Autonomy Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	payload_data_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Thruster Actuation Kinematics	Does Thruster Actuation Kinematics have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	actuation_kinematics	thruster_power_bus; thrust_command_bus; mechanical_mounts; hydrodynamic_shell; alarm_bus; service_access	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
module_allocation	major	ready	Autonomy Compute Communication	Does Autonomy Compute Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	navigation_sensor_bus; payload_data_bus; thrust_command_bus; acoustic_link; alarm_bus; service_access; battery_monitor_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
module_allocation	major	ready	Subsea Energy Storage	Does Subsea Energy Storage have the right responsibility boundary, submodule split and	Named engineering reviewer sign-off, change request or deferral note against the	energy_storage_source	dc_power_bus; battery_monitor_bus; service_access; thermal_path; alarm_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				interface ownership for this concept?	module allocation.				
module_all ocation	major	ready	Pressure Thermal Buoyancy Interface	Does Pressure Thermal Buoyancy Interface have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environme ntal_interf ace	pressure_ boundary; thermal_p ath; buoyancy_ trim; service_ac cess; hydrodyna mic_shell; mechanical_ mounts	auv_depth _rating; auv_press ure_power _recovery _safety	none
module_all ocation	major	ready	Deck Recovery And Service	Does Deck Recovery And Service have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenan ce_service ability	service_ac cess; mechanical_ mounts; pressure_ boundary; dc_power _bus; hydrodyna mic_shell	none	none
module_all ocation	major	ready	Subsea Power Distribution	Does Subsea Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_dis tribution	dc_power _bus; thruster_p ower_bus; payload_p ower_bus; low_voltag e_rail; service_ac cess; alarm_bus	auv_endur ance_targ et; auv_navig ation_cont rol_closur e; auv_press ure_power _recovery _safety	none
module_all ocation	major	ready	Leak Abort And Recovery Safety	Does Leak Abort And Recovery Safety have the right responsibility boundary, submodule split and	Named engineering reviewer sign-off, change request or deferral note against the	safety_pro tection	leak_alarm _bus; alarm_bus ; pressure_ boundary; dc_power _bus; service_ac cess;	auv_depth _rating; auv_press ure_power _recovery _safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				interface ownership for this concept?	module allocation.		acoustic_li nk		
module_all ocat ion	major	ready	Navigation And Payload Sensing	Does Navigation And Payload Sensing have the right responsibi lity boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumenta tion	navigation _sensor_b us; payload_d ata_bus; pressure_ boundary; acoustic_li nk; service_ac cess; alarm_bus	none	none
module_all ocat ion	major	ready	Pressure Structure Containme nt	Does Pressure Structure Containme nt have the right responsibi lity boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	pressure_ boundary; mechanical_mounts; service_ac cess; hydrodyna mic_shell	auv_depth _rating; auv_press ure_power _recovery _safety	none
submodul e_allocatio n	major	ready	Thruster Actuation Kinematic s / Drive electronic s	Do the componen t candidate s and interfaces assigned to Drive electronic s make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	thruster_p ower_bus; thrust_co mmand_b us; alarm_bus	auv_endur ance_targ et; auv_navig ation_cont rol_closur e; auv_press ure_power _recovery _safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodul e_allocation	major	ready	Thruster Actuation Kinematics / Thruster array	Do the component candidate s and interfaces assigned to Thruster array make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	actuation_kinematics	thruster_power_bus; thrust_command_bus; mechanical_mounts	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodul e_allocation	major	ready	Thruster Actuation Kinematics / Trim and stability surfaces	Do the component candidate s and interfaces assigned to Trim and stability surfaces make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	actuation_kinematics	hydrodynamic_shell; mechanical_mounts; service_access	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodul e_allocation	major	ready	Autonomy Compute Communication / Acoustic communications	Do the component candidate s and interfaces assigned to Acoustic communications make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	acoustic_link; service_access; alarm_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodul e_allocation	major	ready	Autonomy Compute Communication / Mission computer stack	Do the component candidate s and interfaces assigned to Mission computer stack make	Reviewer note confirming the submodule role, component allocation and any missing	control_compute_communication	navigation_sensor_bus; payload_data_bus; thrust_command_bus; alarm_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Autonomy Compute Communication / Software health management	engineering sense for its stated purpose?	carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Software health management make engineering sense for its stated purpose?	Reviewer note confirming the submodule e role, component allocation and any missing carriers or capacity checks.	control_compute_communication	alarm_bus ; battery_monitor_bus ; service_access	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodule_allocation	major	ready	Subsea Energy Storage / Charging interface	Do the component candidate s and interfaces assigned to Charging interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule e role, component allocation and any missing carriers or capacity checks.	energy_storage_source	service_access; dc_power_bus; battery_monitor_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodule_allocation	major	ready	Subsea Energy Storage / Energy monitoring	Do the component candidate s and interfaces assigned to Energy monitoring make engineering sense for its stated purpose?	Reviewer note confirming the submodule e role, component allocation and any missing carriers or capacity checks.	energy_storage_source	battery_monitor_bus ; alarm_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_power_recovery_safety	none
submodule_allocation	major	ready	Subsea Energy Storage / Subsea battery pack	Do the component candidate s and interfaces assigned to Subsea	Reviewer note confirming the submodule e role, component	energy_storage_source	dc_power_bus; battery_monitor_bus ; thermal_path	auv_endurance_target; auv_navigation_control_closure; auv_press	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Pressure Thermal Buoyancy Interface / Buoyancy and trim	battery pack make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.	environmental_interface	buoyancy_trim; hydrodynamic_shell; service_access	ure_power_recovery_safety	none
				Do the component candidate s and interfaces assigned to Buoyancy and trim make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.			auv_depth_rating; auv_pressure_recovery_safety	
				Do the component candidate s and interfaces assigned to Pressure boundary management make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.			pressure_boundary; service_access	
				Do the component candidate s and interfaces assigned to Thermal path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.			thermal_path; pressure_boundary; mechanical_mounts	
				Do the component candidate s and interfaces assigned to Deck Recovery And Service / Deck recovery	Reviewer note confirming the submodule role, component			mechanical_mounts; service_access	

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Deck Recovery And Service / Field transport	to Deck recovery make engineering sense for its stated purpose?	t allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Field transport make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; hydrodynamic_shell	none	none
submodule_allocation	major	ready	Deck Recovery And Service / Service access	Do the component candidate s and interfaces assigned to Service access make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.		service_access; pressure_boundary; dc_power_bus	none	none
submodule_allocation	major	ready	Subsea Power Distribution / DC distribution	Do the component candidate s and interfaces assigned to DC distribution make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	dc_power_bus; thruster_power_bus; payload_power_bus	auv_endurance_target; auv_navigation_control_closure; auv_pressure_recovery_safety	none
submodule_allocation	major	ready	Subsea Power Distribution / Isolation and grounding	Do the component candidate s and interfaces assigned to	Reviewer note confirming the submodule role, component	power_distribution	dc_power_bus; alarm_bus ; service_access	auv_endurance_target; auv_navigation_control_closure; auv_press	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				Isolation and grounding make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.			ure_power _recovery _safety	
submodul e_allocation	major	ready	Subsea Power Distribution / Low-voltage rails	Do the componen t candidate s and interfaces assigned to Low-voltage rails make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	power_dis tribution	low_voltag e_rail; payload_p ower_bus; service_ac cess	auv_endur ance_targ et; auv_navig ation_cont rol_closur e; auv_press ure_power _recovery _safety	none
submodul e_allocation	major	ready	Leak Abort And Recovery Safety / Abort and recovery	Do the componen t candidate s and interfaces assigned to Abort and recovery make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	safety_pro tection	alarm_bus ; service_ac cess; acoustic_li nk	auv_depth _rating; auv_press ure_power _recovery _safety	none
submodul e_allocation	major	ready	Leak Abort And Recovery Safety / Leak detection	Do the componen t candidate s and interfaces assigned to Leak detection make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	safety_pro tection	leak_alarm _bus; alarm_bus ; pressure_ boundary	auv_depth _rating; auv_press ure_power _recovery _safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Leak Abort And Recovery Safety / Pressure fail-safe	Do the component candidates and interfaces assigned to Pressure fail-safe make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	pressure_boundary; dc_power_bus; alarm_bus	auv_depth_rating; auv_pressure_power_recovery_safety	none

Showing first 70 of 76 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Depth Rating	pass	300 m	10 to 600 m for inspection-class AUV prototypes; deeper missions need dedicated pressure-vessel, seal and recovery evidence	300 m is plausible for an inspection AUV before detailed pressure-boundary calculation, material selection and proof testing.
Survey Endurance Target	pass	8 hours	<=12 h plausible for compact inspection AUV concept, 12-24 h needs battery/drag budget, >24 h is high risk without a detailed energy model	8 hours is credible only after drag, hotel-load, payload-load and reserve-energy calculations are completed.
Navigation/Control Closure	pass	present	navigation sensor bus, payload data bus, acoustic link and thrust command bus present	DVL/INS/depth sensing, payload data, acoustic messages and thruster commands are connected before BoM sourcing.
Pressure/Power/Recovery Safety	pass	present	pressure boundary, leak alarm, alarm bus, DC power, service access and recovery beacon present	The architecture names the hull boundary, leak detection, power isolation and recovery mechanisms needed before sourcing components.

Engineering Calculation Ledger

2 calculations: 1 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
AUV hydrostatic pressure	within envelope	1 + depth_rating_m / 10	31 bar abs	<=51 bar abs covers up to roughly 500 m before deeper-hull concept review.	31 bar absolute pressure implied at rated depth.	Confirm hull material, safety factor, penetrators, seals, buckling margin and test pressure.
AUV endurance minutes	needs review	endurance_hours * 60	480 minutes	Arithmetic only; mission endurance needs hotel load, propulsion and reserve energy proof.	480 minutes mission endurance before energy-budget proof.	Confirm speed profile, payload load, reserve margin, battery derating and recovery scenario.

Engineering Assumption Ledger

32 assumptions: 2 brief-supported, 8 model-present, 8 needing engineering review, 14 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 14.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Endurance	brief_requirement	brief supported	no	no	Endurance = 8 hours is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Depth rating	brief_requirement	brief supported	no	no	Depth rating = 300 m is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Survey endurance	derived_metric	review required	no	no	Survey endurance uses endurance_hours with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Depth Rating	sanity_envelope	review required	no	no	Depth Rating currently evaluates as pass against 10 to 600 m for inspection-class AUV prototypes; deeper missions need dedicated pressure-vessel, seal and recovery evidence.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Survey Endurance Target	sanity_envelope	review required	no	no	Survey Endurance Target currently evaluates as pass against <=12 h plausible for compact inspection AUV	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					concept, 12-24 h needs battery/drag budget, >24 h is high risk without a detailed energy model.	for this envelope.
Navigation/Control Closure	sanity_envelope	review required	no	no	Navigation/Control Closure currently evaluates as pass against navigation sensor bus, payload data bus, acoustic link and thrust command bus present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Pressure/Power/Recovery Safety	sanity_envelope	review required	no	no	Pressure/Power/Recovery Safety currently evaluates as pass against pressure boundary, leak alarm, alarm bus, DC power, service access and recovery beacon present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
energy_storage_source -> power_distribution	interface_closure	model present	no	no	dc_power_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
power_distribution -> actuation_kinematics	interface_closure	model present	no	no	thruster_power_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> actuation_kinematics	interface_closure	model present	no	no	thrust_command_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
sensing_instrumentation -> control_computer_communication	interface_closure	model present	no	no	navigation_sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_computer_communication	interface_closure	model present	no	no	payload_data_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
environmental_interface -> structure_containment	interface_closure	model present	no	no	pressure_boundary is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
safety_protection -> control_computer_communication	interface_closure	model present	no	no	alarm_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
safety_protection -> structure_containment	interface_closure	model present	no	no	pressure_boundary is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
Pressure hull	critical_component	source required	no	yes	Pressure hull is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Battery pack	critical_component	source required	no	yes	Battery pack is a critical architecture-derived	Admit supplier, manufacturer, MPN where applicable, unit

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					component candidate with quantity 1 candidate.	cost, lead time and evidence reference through sourcing intake.
Wet-mate connector set	critical_component	source required	no	yes	Wet-mate connector set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Thruster set	critical_component	source required	no	yes	Thruster set is a critical architecture-derived component candidate with quantity 6 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Mission computer	critical_component	source required	no	yes	Mission computer is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Acoustic modem	critical_component	source required	no	yes	Acoustic modem is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
DVL	critical_component	source required	no	yes	DVL is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Inertial navigation unit	critical_component	source required	no	yes	Inertial navigation unit is a critical architecture-derived component candidate with	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					quantity 1 candidate.	through sourcing intake.
Depth and pressure sensor	critical_component	source required	no	yes	Depth and pressure sensor is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Forward sonar payload	critical_component	source required	no	yes	Forward sonar payload is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Leak detection sensor	critical_component	source required	no	yes	Leak detection sensor is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Recovery beacon	critical_component	source required	no	yes	Recovery beacon is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
PRESSURE-BOUNDARY-REVIEW	compliance_evidence	review required	no	no	Depth-rated pressure boundary review is relevant to this product class in Project engineering gate.	Depth rating calculation, hull material allowables, endcap seal design and proof-pressure plan.
SUBSEA-LEAK-ABORT-REVIEW	compliance_evidence	review required	no	no	Leak detection and abort-state review is relevant to this product class in Project engineering gate.	Leak sensor placement, abort state machine, battery isolation and recovery procedure.
EMC-MARINE-COMMS-REVIEW	compliance_evidence	review required	no	no	Marine communications and EMC review	Acoustic modem integration, antenna/transdu

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					is relevant to this product class in Project engineering gate.	cer mounting, cable shielding and data-loss behaviour.

Engineering Verification Plan

37 activities: 24 ready for review, 6 open, 7 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Pressure Structure Containment architecture allocation	ready	design_review	Pressure Structure Containment	depth_rating_m	pressure_bondary; mechanical_mounts; service_access; hydrodynamic_shell	pressure_hull; endcap_clamp_ring; double_oring_gland_set; hull_proof_test_port; electronics_tray_rail; battery_cradle_frame; payload_mounting_rail; trim_weight_pocket; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Pressure Structure Containment requirement calculations	ready	calculation	Pressure Structure Containment	depth_rating_m	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Pressure Structure Containment critical components	blocked	source_evidence	Pressure Structure Containment	depth_rating_m	none	pressure_hull	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	pressure_hull

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Subsea Energy Storage architecture allocation	ready	design_review	Subsea Energy Storage	endurance_hours	dc_power_bus; battery_monitor_bus; service_access; thermal_path; alarm_bus	battery_pack; battery_management_board; pack_isolation_fuse; pressure_rated_battery_enclosure; charge_interlock_plug; shore_charger_adapter; state_of_charge_display_puck; battery_vent_inspection_plug; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Subsea Energy Storage requirement calculations	ready	calculation	Subsea Energy Storage	endurance_hours	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Subsea Energy Storage critical components	blocked	source_evidence	Subsea Energy Storage	endurance_hours	none	battery_pack	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	battery_pack
Review Subsea Power Distribution architecture allocation	ready	design_review	Subsea Power Distribution	endurance_hours	dc_power_bus; thruster_power_bus; payload_power_bus; low_voltage_rail; service_access; alarm_bus	fused_dc_distribution_board; wet_mate_connector_set; main_power_enable_relay; bus_voltage_monitor; 24_v_dc_converter; 12_v_payload_converter; 5_v_avionics_converter; emi_power_filter; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Subsea Power Distribution requirement calculations	ready	calculation	Subsea Power Distribution	endurance_hours	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Subsea Power Distribution critical components	blocked	source_evidence	Subsea Power Distribution	endurance_hours	none	wet_mate_connector_set	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	wet_mate_connector_set
Review Thruster Actuation Kinematics architecture allocation	ready	design_review	Thruster Actuation Kinematics	endurance_hours	thruster_power_bus; thrust_command_bus; mechanical_mounts; hydrodynamic_shell; alarm_bus; service_access	thruster_set; thruster_nozzle_guard; thruster_mounting_bracket; motor_phase_harness; thruster_motor_controller_stack; current_sensor_channel_set; controller_heat_spreader; thrust_calibration_table	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						; plus 4 more	No blocker or major architecture issue is attached to the module.	
Verify Thruster Actuation Kinematics requirement calculations	ready	calculation	Thruster Actuation Kinematics	endurance_hours	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Thruster Actuation Kinematics critical components	blocked	source_evidence	Thruster Actuation Kinematics	endurance_hours	none	thruster_set	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	thruster_set
Review Autonomy Compute Communication	ready	design_review	Autonomy Compute Communication	endurance_hours	navigation_sensor_bus; payload_data_bus; thrust_command_bus;	mission_computer; real_time_control_board; watchdog_s	Every submodule has a stated purpose and at least one	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
architecture allocation					acoustic_link; alarm_bus; service_access; battery_monitor_bus	upervisor_module; mission_data_recorder; acoustic_modem; acoustic_transducer_module; acoustic_isolation_pad; surface_command_protocol_adapter; plus 4 more	component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Autonomy Compute Communication requirement calculations	ready	calculation	Autonomy Compute Communication	endurance_hours	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Autonomy Compute Communication critical components	blocked	source_evidence	Autonomy Compute Communication	endurance_hours	none	mission_computer; acoustic_modem	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed	mission_computer acoustic_modem

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							quantity and component identity match the architecture candidate line.	
Review Navigation And Payload Sensing architecture allocation	ready	design_review	Navigation And Payload Sensing	none	navigation_sensor_bus; payload_data_bus; pressure_bondary; acoustic_link; service_access; alarm_bus	dvl; inertial_navigation_unit; magnetic_compass_module; gnss_surface_receiver; depth_and_pressure_sensor; water_temperature_probe; conductivity_sensor; internal_humidity_sensor; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Navigation And Payload Sensing critical components	blocked	source_evidence	Navigation And Payload Sensing	none	none	dvl; inertial_navigation_unit; depth_and_pressure_sensor; forward_sonar_payload	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	dvl inertial_navigation_unit depth_and_pressure_sensor forward_sonar_payload

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Pressure Thermal Buoyancy Interface architecture allocation	ready	design_review	Pressure Thermal Buoyancy Interface	depth_rating_m	pressure_boundary; thermal_patching; buoyancy_trim; service_access; hydrodynamic_shell; mechanical_mounts	cable_penet_rator_seal; pressure_compensation_bladder; oil_fill_service_port; pressure_equalisation_valve; electronics_thermal_bridge; hull_heat_spreader_plate; battery_thermal_pad; thruster_heat_sink_collar; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Pressure Thermal Buoyancy Interface requirement calculations	ready	calculation	Pressure Thermal Buoyancy Interface	depth_rating_m	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Leak Abort And Recovery Safety architecture allocation	ready	design_review	Leak Abort And Recovery Safety	depth_rating_m	leak_alarm_bus; alarm_bus; pressure_boundary; dc_power_bus; service_access; acoustic_link	leak_detection_sensor; bilge_moisture_strip; leak_alarm_relay; absorbent_witness_padded; recovery_beacon; drop_weight_release; emergency_strobe;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						surface_float; plus 4 more	where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Leak Abort And Recovery Safety requirement calculations	ready	calculation	Leak Abort And Recovery Safety	depth_rating_m	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Leak Abort And Recovery Safety critical components	blocked	source_evidence	Leak Abort And Recovery Safety	depth_rating_m	none	leak_detection_sensor; recovery_beacon	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	leak_detection_sensor recovery_beacon

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Deck Recovery And Service architecture allocation	ready	design_review	Deck Recovery And Service	none	service_access; mechanical_mounts; pressure_bondary; dc_power_buss; hydrodynamic_shell	lifting_bridle; deck_recovery_handle; launch_cradle_interface; tow_point_fitting; o_ring_grease_kit; vacuum_leak_test_port; diagnostic_connector; service_checklist_card; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify dc_power_buss interface contract	ready	interface_review	Subsea Energy Storage -> Subsea Power Distribution	none	dc_power_buss	battery_pack; battery_management_board; pack_isolation_fuse; pressure_rated_battery_enclosure; charge_intelock_plug; shore_charger_adapter; state_of_charge_display_puck; battery_vent_inspection_plug; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify thruster_power_bus interface contract	ready	interface_review	Subsea Power Distribution -> Thruster Actuation Kinematics	none	thruster_power_bus	fused_dc_distribution_board; wet_mate_connector_set; main_power_enable_relay; bus_voltage_monitor; thruster_set; thruster_no	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid,	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						zzle_guard; thruster_mounting_bracket; motor_phase_harness; plus 4 more	mechanical or data capacity is validated by the next detailed design pass.	
Verify thrust_command_bus interface contract	ready	interface_review	Autonomy Compute Communication -> Thruster Actuation Kinematics	none	thrust_command_bus	mission_computer; real_time_control_board; watchdog_supervisor_module; mission_data_recorder; thruster_set; thruster_nozzle_guard; thruster_mounting_bracket; motor_phase_harness; plus 4 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify navigation_sensor_bus interface contract	ready	interface_review	Navigation And Payload Sensing -> Autonomy Compute Communication	none	navigation_sensor_bus	dvl; inertial_navigation_unit; magnetic_compass_module; gnss_surface_receiver; depth_and_pressure_sensor; water_temperature_probe; conductivity_sensor; internal_humidity_sensor; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify payload_data_bus interface contract	ready	interface_review	Navigation And Payload Sensing -> Autonomy Compute Communication	none	payload_data_bus	forward_sonar_payload; camera_viewport_window; payload_sync_trigger; payload_data_link_module;	Both endpoint modules declare the interface. At least one submodule carrier is named on each	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						mission_computer; real_time_control_board; watchdog_supervisor_module; mission_data_recorder	endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify pressure_boudary interface contract	ready	interface_review	Pressure Thermal Buoyancy Interface -> Pressure Structure Containment	none	pressure_boudary	cable_penet_rator_seal; pressure_compensation_bladder; oil_fill_service_port; pressure_equalisation_valve; electronics_thermal_bridge; hull_heat_spreader_plate; battery_thermal_pad; thruster_heat_sink_collar; plus 4 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify alarm_bus interface contract	ready	interface_review	Leak Abort And Recovery Safety -> Autonomy Compute Communication	none	alarm_bus	leak_detection_sensor; bilge_moisture_strip; leak_alarm_relay; absorbent_witness_pad; recovery_beacon; drop_weight_release; emergency_strobe; surface_floating_float; plus 16 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify pressure_boudary interface contract	ready	interface_review	Leak Abort And Recovery Safety -> Pressure Structure	none	pressure_boudary	leak_detection_sensor; bilge_moisture_strip; leak_alarm_relay;	Both endpoint modules declare the interface. At least one	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
			Containment			absorbent_witness_padded; over_depth_abort_switch; watchdog_kill_relay; battery_isolation_constructor; pressure_relief_service_plug; plus 4 more	submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Plan evidence for PRESSURE-BOUNDARY-REVIEW	open	compliance_review	System Compliance	none	none	none	Depth rating calculation, hull material allowables, endcap seal design and proof-pressure plan.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for SUBSEA-LEAK-ABORT-REVIEW	open	compliance_review	System Compliance	none	none	none	Leak sensor placement, abort state machine, battery isolation and recovery procedure.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for EMC-MARINE-COMMS-REVIEW	open	compliance_review	System Compliance	none	none	none	Acoustic modem integration, antenna/transducer mounting, cable shielding and data-loss behaviour.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Pressure hull or seal failure at depth	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							residual high-risk item has an owner before publication.	
Review risk control for Loss of navigation or communications underwater	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for Unable to recover vehicle after abort	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 30 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Pressure Structure Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Pressure Structure Containment requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Pressure Structure Containment critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Subsea Energy Storage architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Subsea Energy Storage requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Subsea Energy Storage critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Subsea Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Subsea Power Distribution requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Subsea Power Distribution critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Thruster Actuation Kinematics architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Thruster Actuation Kinematics requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Thruster Actuation	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN,

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Kinematics critical components					cost and lead-time evidence.
Review Autonomy Compute Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Autonomy Compute Communication requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Autonomy Compute Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Navigation And Payload Sensing architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Navigation And Payload Sensing critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Pressure Thermal Buoyancy Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Pressure Thermal Buoyancy Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Leak Abort And Recovery Safety architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Leak Abort And Recovery Safety requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Leak Abort And Recovery Safety critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Deck Recovery And Service architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify dc_power_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify thruster_power_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify thrust_command_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify navigation_sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify payload_data_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify pressure_boundary interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify alarm_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify pressure_boundary interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for PRESSURE-BOUNDARY-REVIEW	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for SUBSEA-LEAK-ABORT-REVIEW	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for EMC-MARINE-COMMS-REVIEW	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Pressure hull or seal failure at depth	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review risk control for Loss of navigation or communications underwater	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Unable to recover vehicle after abort	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Pressure Structure Containment

Maintains a dry electronics and battery envelope for a 300 m underwater operating target.

- pressure_boundary
- mechanical_mounts
- service_access
- hydrodynamic_shell

Pressure hull shell

Forms the dry pressure boundary and endcap sealing envelope.

Components: Pressure hull, endcap clamp ring, double O-ring gland set, hull proof-test port

Internal mounting frame

Locates dry electronics, batteries and payload interface hardware.

Components: electronics tray rail, battery cradle frame, payload mounting rail, trim weight pocket

Hydrodynamic fairing

Reduces drag and protects external appendages during survey runs.

Components: nose fairing, tail cone fairing, stabilising fin set, access hatch latch set

Subsea Energy Storage

Stores enough mission energy for roughly 8 hours before detailed drag, hotel-load and reserve calculations.

- dc_power_bus
- battery_monitor_bus
- service_access
- thermal_path

Subsea battery pack

Supplies mission energy with reserve margin and pressure-rated containment.

Components: Battery pack, battery management board, pack isolation fuse, pressure-rated battery enclosure

Charging interface

Enables deck charging and state verification without compromising the pressure boundary.

Components: charge interlock plug, shore charger adapter, state-of-charge display puck, battery vent inspection plug

Energy monitoring

Tracks reserve energy and supports abort decisions.

Components: coulomb counter module, pack temperature sensor strip, energy reserve indicator, battery telemetry isolator

Subsea Power Distribution

Routes protected power to thrusters, mission computer, navigation sensors and payloads.

dc_power_bus	thruster_power_bus	payload_power_bus	low_voltage_rail	service_access
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DC distribution

Switches and protects the main battery bus before branch loads.

Components: fused DC distribution board, Wet-mate connector set, main power enable relay, bus voltage monitor

Low-voltage rails

Creates stable avionics and payload supply rails.

Components: 24 V DC converter, 12 V payload converter, 5 V avionics converter, EMI power filter

Isolation and grounding

Controls leakage, bonding and service isolation paths in a flooded environment.

Components: ground fault monitor, chassis bonding strap, service isolation key, fault annunciator module

Thruster Actuation Kinematics

Generates controlled surge, heave, yaw and station-keeping authority underwater.

thruster_power_bus	thrust_command_bus	mechanical_mounts	hydrodynamic_shell
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Thruster array

Provides vectored propulsion channels for survey and manoeuvre control.

Components: Thruster set, thruster nozzle guard, thruster mounting bracket, motor phase harness

Drive electronics

Converts command inputs into motor phase drive while reporting current and faults.

Components: thruster motor controller stack, current sensor channel set, controller heat spreader, thrust calibration table

Trim and stability surfaces

Improves passive stability and reduces control effort.

Components: adjustable trim fin, ballast trim block, roll trim screw set, tow-tank trim mark

Autonomy Compute Communication

Runs mission autonomy, navigation fusion, acoustic messaging, payload logging and fail-safe states.

navigation_sensor_bus	payload_data_bus	thrust_command_bus	acoustic_link	alarm_bus	service_access
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Mission computer stack

Executes mission state machine, vehicle control and payload logging.

Components: Mission computer, real-time control board, watchdog supervisor module, mission data recorder

Acoustic communications

Provides low-bandwidth underwater command, health and recovery messaging.

Components: Acoustic modem, acoustic transducer mount, acoustic isolation pad, surface command protocol adapter

Software health management

Defines safe modes, reserve-energy logic and post-mission evidence.

Components: failsafe state machine, time sync module, configuration backup storage, debug service port

Navigation And Payload Sensing

Measures vehicle state, seabed-relative movement, depth and inspection payload data.

navigation_sensor_bus	payload_data_bus	pressure_boundary	acoustic_link
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Navigation sensor suite

Combines bottom-track velocity, inertial attitude and surface fixes.

Components: DVL, Inertial navigation unit, magnetic compass module, GNSS surface receiver

Depth and environment sensors

Measures depth, water conditions and pressure-boundary state.

Components: Depth and pressure sensor, water temperature probe, conductivity sensor, internal humidity sensor

Payload sensor bay

Carries inspection payloads and synchronises payload data with navigation state.

Components: Forward sonar payload, camera viewport window, payload sync trigger, payload data link module

Pressure Thermal Buoyancy Interface

Manages pressure boundary support, heat rejection, buoyancy and trim interactions with seawater.

pressure_boundary	thermal_path	buoyancy_trim	service_access	hydrodynamic_shell
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Pressure boundary management

Handles penetrators, pressure compensation and pressure-test interfaces.

Components: cable penetrator seal, pressure compensation bladder, oil-fill service port, pressure equalisation valve

Thermal path

Conducts electronics and battery heat into the hull and surrounding seawater.

Components: electronics thermal bridge, hull heat spreader plate, battery thermal pad, thruster heat sink collar

Buoyancy and trim

Sets neutral buoyancy, trim and recoverable mass distribution.

Components: syntactic foam block set, ballast rail, buoyancy calibration tag, field trim worksheet

Leak Abort And Recovery Safety

Detects water ingress, over-depth risk and lost-vehicle states, then drives abort and recovery behaviour.

leak_alarm_bus	alarm_bus	pressure_boundary	dc_power_bus	service_access
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Leak detection

Detects internal water ingress before catastrophic electronics damage.

Components: Leak detection sensor, bilge moisture strip, leak alarm relay, absorbent witness pad

Abort and recovery

Supports vehicle location and recovery after fault or mission completion.

Components: Recovery beacon, drop-weight release, emergency strobe, surface flag float

Pressure fail-safe

Prevents over-depth operation and isolates hazardous energy under fault.

Components: over-depth abort switch, watchdog kill relay, battery isolation contactor, pressure relief service plug

Deck Recovery And Service

Supports launch, recovery, charging, leak testing and field service without disturbing verified settings unnecessarily.

- service_access
- mechanical_mounts
- pressure_boundary
- dc_power_bus

Deck recovery

Provides safe handling points for launch and retrieval operations.

Components: lifting bridle, deck recovery handle, launch cradle interface, tow point fitting

Service access

Enables repeatable checks for seals, charging and diagnostics.

Components: O-ring grease kit, vacuum leak test port, diagnostic connector, service checklist card

Field transport

Protects the vehicle and records environmental exposure between missions.

Components: transport cradle, protective nose cover, desiccant cartridge, maintenance log tag

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Pressure hull	1	candidate	unpriced	unpriced				
endcap clamp ring	1	candidate	unpriced	unpriced				
double O-ring gland set	1	candidate	unpriced	unpriced				
hull proof-test port	1	candidate	unpriced	unpriced				
electronics tray rail	1	candidate	unpriced	unpriced				
battery cradle frame	1	candidate	unpriced	unpriced				
payload mounting rail	1	candidate	unpriced	unpriced				
trim weight pocket	1	candidate	unpriced	unpriced				
nose fairing	1	candidate	unpriced	unpriced				
tail cone fairing	1	candidate	unpriced	unpriced				
stabilising fin set	1	candidate	unpriced	unpriced				
access hatch latch set	1	candidate	unpriced	unpriced				
Battery pack	1	candidate	unpriced	unpriced				
battery management board	1	candidate	unpriced	unpriced				
pack isolation fuse	1	candidate	unpriced	unpriced				
pressure-rated battery enclosure	1	candidate	unpriced	unpriced				
charge interlock plug	1	candidate	unpriced	unpriced				
shore charger adapter	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
state-of-charge display puck	1	candidate	unpriced	unpriced				
battery vent inspection plug	1	candidate	unpriced	unpriced				
coulomb counter module	1	candidate	unpriced	unpriced				
pack temperature sensor strip	1	candidate	unpriced	unpriced				
energy reserve indicator	1	candidate	unpriced	unpriced				
battery telemetry isolator	1	candidate	unpriced	unpriced				
fused DC distribution board	1	candidate	unpriced	unpriced				
Wet-mate connector set	1	candidate	unpriced	unpriced				
main power enable relay	1	candidate	unpriced	unpriced				
bus voltage monitor	1	candidate	unpriced	unpriced				
24 V DC converter	1	candidate	unpriced	unpriced				
12 V payload converter	1	candidate	unpriced	unpriced				
5 V avionics converter	1	candidate	unpriced	unpriced				
EMI power filter	1	candidate	unpriced	unpriced				
ground fault monitor	1	candidate	unpriced	unpriced				
chassis bonding strap	1	candidate	unpriced	unpriced				
service isolation key	1	candidate	unpriced	unpriced				
fault annunciator	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
module								
Thruster set	6	each	unpriced	unpriced				
thruster nozzle guard	1	candidate	unpriced	unpriced				
thruster mounting bracket	1	candidate	unpriced	unpriced				
motor phase harness	1	candidate	unpriced	unpriced				
thruster motor controller stack	1	candidate	unpriced	unpriced				
current sensor channel set	1	candidate	unpriced	unpriced				
controller heat spreader	1	candidate	unpriced	unpriced				
thrust calibration table	1	candidate	unpriced	unpriced				
adjustable trim fin	1	candidate	unpriced	unpriced				
ballast trim block	1	candidate	unpriced	unpriced				
roll trim screw set	1	candidate	unpriced	unpriced				
tow-tank trim mark	1	candidate	unpriced	unpriced				
Mission computer	1	candidate	unpriced	unpriced				
real-time control board	1	candidate	unpriced	unpriced				
watchdog supervisor module	1	candidate	unpriced	unpriced				
mission data recorder	1	candidate	unpriced	unpriced				
Acoustic modem	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
acoustic transducer mount	1	candidate	unpriced	unpriced				
acoustic isolation pad	1	candidate	unpriced	unpriced				
surface command protocol adapter	1	candidate	unpriced	unpriced				
failsafe state machine	1	candidate	unpriced	unpriced				
time sync module	1	candidate	unpriced	unpriced				
configuration backup storage	1	candidate	unpriced	unpriced				
debug service port	1	candidate	unpriced	unpriced				
DVL	1	candidate	unpriced	unpriced				
Inertial navigation unit	1	candidate	unpriced	unpriced				
magnetic compass module	1	candidate	unpriced	unpriced				
GNSS surface receiver	1	candidate	unpriced	unpriced				
Depth and pressure sensor	1	candidate	unpriced	unpriced				
water temperature probe	1	candidate	unpriced	unpriced				
conductivity sensor	1	candidate	unpriced	unpriced				
internal humidity sensor	1	candidate	unpriced	unpriced				
Forward sonar payload	1	candidate	unpriced	unpriced				
camera viewport window	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
payload sync trigger	1	candidate	unpriced	unpriced				
payload data link module	1	candidate	unpriced	unpriced				
cable penetrator seal	1	candidate	unpriced	unpriced				
pressure compensation bladder	1	candidate	unpriced	unpriced				
oil-fill service port	1	candidate	unpriced	unpriced				
pressure equalisation valve	1	candidate	unpriced	unpriced				
electronics thermal bridge	1	candidate	unpriced	unpriced				
hull heat spreader plate	1	candidate	unpriced	unpriced				
battery thermal pad	1	candidate	unpriced	unpriced				
thruster heat sink collar	1	candidate	unpriced	unpriced				
syntactic foam block set	1	candidate	unpriced	unpriced				
ballast rail	1	candidate	unpriced	unpriced				
buoyancy calibration tag	1	candidate	unpriced	unpriced				
field trim worksheet	1	candidate	unpriced	unpriced				
Leak detection sensor	1	candidate	unpriced	unpriced				
bilge moisture strip	1	candidate	unpriced	unpriced				
leak alarm relay	1	candidate	unpriced	unpriced				
absorbent witness pad	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Recovery beacon	1	candidate	unpriced	unpriced				
drop-weight release	1	candidate	unpriced	unpriced				
emergency strobe	1	candidate	unpriced	unpriced				
surface flag float	1	candidate	unpriced	unpriced				
over-depth abort switch	1	candidate	unpriced	unpriced				
watchdog kill relay	1	candidate	unpriced	unpriced				
battery isolation contactor	1	candidate	unpriced	unpriced				
pressure relief service plug	1	candidate	unpriced	unpriced				
lifting bridle	1	candidate	unpriced	unpriced				
deck recovery handle	1	candidate	unpriced	unpriced				
launch cradle interface	1	candidate	unpriced	unpriced				
tow point fitting	1	candidate	unpriced	unpriced				
O-ring grease kit	1	candidate	unpriced	unpriced				
vacuum leak test port	1	candidate	unpriced	unpriced				
diagnostic connector	1	candidate	unpriced	unpriced				
service checklist card	1	candidate	unpriced	unpriced				
transport cradle	1	candidate	unpriced	unpriced				
protective nose cover	1	candidate	unpriced	unpriced				
desiccant cartridge	1	candidate	unpriced	unpriced				
maintenance log tag	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	108	0	108	12

Critical Sourcing Worklist

Component	Qty	Reason
Pressure hull	1 candidate	Required class-critical line must be sourced before BoM can pass.
Battery pack	1 candidate	Required class-critical line must be sourced before BoM can pass.
Wet-mate connector set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Thruster set	6 each	Required class-critical line must be sourced before BoM can pass.
Mission computer	1 candidate	Required class-critical line must be sourced before BoM can pass.
Acoustic modem	1 candidate	Required class-critical line must be sourced before BoM can pass.
DVL	1 candidate	Required class-critical line must be sourced before BoM can pass.
Inertial navigation unit	1 candidate	Required class-critical line must be sourced before BoM can pass.
Depth and pressure sensor	1 candidate	Required class-critical line must be sourced before BoM can pass.
Forward sonar payload	1 candidate	Required class-critical line must be sourced before BoM can pass.
Leak detection sensor	1 candidate	Required class-critical line must be sourced before BoM can pass.
Recovery beacon	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/108 priced lines admitted. Critical coverage 0/12. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
Pressure hull	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
endcap clamp ring	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
double O-ring gland set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
hull proof-test port	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
electronics tray rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery cradle frame	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
payload mounting rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
trim weight pocket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
nose fairing	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
tail cone fairing	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
stabilising fin set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
access hatch latch set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
Battery pack	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
battery management board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
pack isolation fuse	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
pressure-rated battery enclosure	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
charge interlock plug	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
shore charger adapter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
state-of-charge display puck	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery vent inspection plug	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
coulomb counter module	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
pack temperature sensor strip	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
energy reserve indicator	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
battery telemetry isolator	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
fused DC distribution board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Wet-mate connector set	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
main power enable relay	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
bus voltage monitor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
24 V DC converter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
12 V payload converter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
5 V avionics converter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
EMI power filter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
ground fault monitor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
chassis bonding strap	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
service isolation key	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
fault annunciator module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Thruster set	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
thruster nozzle guard	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
thruster mounting bracket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
motor phase harness	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 108 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 48 missing-source claims, 0 provenance violations. Critical missing-source claims: 48. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
Pressure hull	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Pressure hull	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Pressure hull	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Pressure hull	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Battery pack	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Battery pack	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Battery pack	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Battery pack	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Wet-mate connector set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Wet-mate connector set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Wet-mate connector set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Wet-mate connector set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thruster set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thruster set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thruster set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thruster set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Mission computer	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Mission computer	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Mission computer	mpn	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Mission computer	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Acoustic modem	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Acoustic modem	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Acoustic modem	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Acoustic modem	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DVL	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DVL	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DVL	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DVL	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Inertial navigation unit	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inertial navigation unit	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inertial navigation unit	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inertial navigation unit	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Depth and pressure sensor	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Depth and pressure sensor	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Depth and pressure sensor	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Depth and pressure sensor	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Forward sonar payload	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Forward sonar payload	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Forward sonar payload	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Forward sonar payload	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Leak detection sensor	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Leak detection sensor	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Leak detection sensor	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Leak detection sensor	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery beacon	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery beacon	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery beacon	mpn	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery beacon	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/108 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 12 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_pressure_hull_shell_pressure_hull.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Pressure hull	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
endcap clamp ring	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
double O-ring gland set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
hull proof-test port	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
electronics tray rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery cradle frame	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
payload mounting rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
trim weight pocket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
nose fairing	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
tail cone fairing	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
stabilising fin set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
access hatch latch set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Battery pack	critical un sourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
battery management board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pack isolation fuse	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pressure-rated battery enclosure	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
charge interlock plug	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
shore charger adapter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
state-of-charge display puck	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery vent inspection plug	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
coulomb counter module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pack temperature sensor strip	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									procurement planning starts.
energy reserve indicator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery telemetry isolator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
fused DC distribution board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Wet-mate connector set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
main power enable relay	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
bus voltage monitor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24 V DC converter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
12 V payload converter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
5 V avionics converter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
EMI power filter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ground fault monitor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									sourcing is complete or procurement planning starts.
chassis bonding strap	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
service isolation key	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
fault annunciator module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Thruster set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
thruster nozzle guard	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
thruster mounting bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
motor phase harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
thruster motor controller stack	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
current sensor channel set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
controller heat spreader	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
thrust calibration table	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
adjustable trim fin	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
ballast trim block	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
roll trim screw set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
tow-tank trim mark	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Mission computer	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									for this critical line before BoM can pass.
real-time control board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 108 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

108 closure row(s): 12 ready, 0 blocked, 96 deferred. Actions: 12 collect source, 0 repair, 0 replace protocol, 96 candidate deferrals. Procurement blockers 12; priced-review blockers 12; next row bom-close:actuation_kinematics_thruster_array_thruster_set.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	Thruster set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Acoustic modem	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Mission computer	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	Battery pack	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Wet-mate connector set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Recovery beacon	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Leak detection sensor	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Depth and pressure sensor	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	DVL	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
10	Inertial navigation unit	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
11	Forward sonar payload	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
12	Pressure hull	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
13	controller heat spreader	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	current sensor channel set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	thrust calibration table	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	thruster motor controller stack	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	motor phase harness	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
18	thruster mounting bracket	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	thruster nozzle guard	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	adjustable trim fin	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	ballast trim block	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	roll trim screw set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	tow-tank trim mark	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	acoustic isolation pad	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	acoustic transducer mount	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	surface command protocol adapter	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	mission data recorder	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	real-time control board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	watchdog supervisor module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	configuration backup storage	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	debug service port	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	failsafe state machine	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	time sync module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	battery vent inspection plug	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	charge interlock plug	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	shore charger adapter	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	state-of-charge display puck	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	battery telemetry isolator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	coulomb counter module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	energy reserve indicator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	pack temperature sensor strip	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	battery management board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	pack isolation fuse	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	pressure-rated battery enclosure	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	ballast rail	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	buoyancy calibration tag	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	field trim worksheet	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	syntactic foam block set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	cable penetrator seal	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	oil-fill service port	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 108 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 12; deferred rows 96. Critical source rows 12; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:thruster_set.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Thruster set	Thruster set; Thruster set manufacturer part number; Thruster set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Acoustic modem	Acoustic modem; Acoustic modem manufacturer part number; Acoustic modem catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Mission computer	Mission computer; Mission computer manufacturer part number; Mission computer catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Battery pack	Battery pack; Battery pack manufacturer part number; Battery pack catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Wet-mate connector set	Wet-mate connector set; Wet-mate connector set manufacturer part number; Wet-mate connector set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Recovery beacon	Recovery beacon; Recovery beacon manufacturer part number; Recovery beacon catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Leak detection sensor	Leak detection sensor; Leak detection sensor manufacturer part number; Leak detection sensor catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Depth and pressure sensor	Depth and pressure sensor; Depth and pressure sensor manufacturer part number; Depth and pressure sensor catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	DVL	DVL; DVL manufacturer part number; DVL catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Inertial navigation unit	Inertial navigation unit; Inertial navigation unit manufacturer part number; Inertial navigation unit catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Forward sonar payload	Forward sonar payload; Forward sonar payload manufacturer part number; Forward sonar payload catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Pressure hull	Pressure hull; Pressure hull manufacturer part number; Pressure hull catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	controller heat spreader	controller heat spreader; controller heat spreader manufacturer part number; controller heat spreader catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	current sensor channel set	current sensor channel set; current sensor channel set manufactur	no immediate evidence required unless this candidate is promoted to	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					er part number; current sensor channel set catalogue price GBP	procurement scope		
6	Defer Candidate-Only Sourcing	deferred	minor	thrust calibration table	thrust calibration table; thrust calibration table manufacturer part number; thrust calibration table catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	thruster motor controller stack	thruster motor controller stack; thruster motor controller stack manufacturer part number; thruster motor controller stack catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor phase harness	motor phase harness; motor phase harness manufacturer part number; motor phase harness catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	thruster mounting bracket	thruster mounting bracket; thruster mounting bracket manufacturer part number; thruster mounting bracket catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	thruster nozzle guard	thruster nozzle guard; thruster nozzle guard manufacturer part number; thruster nozzle guard catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	adjustable trim fin	adjustable trim fin; adjustable trim fin manufacturer part number; adjustable trim fin catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	ballast trim block	ballast trim block; ballast trim block manufacturer part number; ballast trim block catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	roll trim screw set	roll trim screw set; roll trim screw set manufacturer part number; roll trim screw set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	tow-tank trim mark	tow-tank trim mark; tow-tank trim mark manufacturer part number; tow-tank trim mark catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	acoustic isolation pad	acoustic isolation pad; acoustic isolation pad manufacturer part number; acoustic isolation pad catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/12; procurement-blocking closure rows 12; active sourcing rows 12; deferred sourcing rows 96. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_review_required. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/12 critical rows production-eligible; 12 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_pressure_hull_s hell_pressure_hull: Collect admissible source evidence for this critical line before BoM can pass. energy_storage_source_subsea_battery_pack_battery_pack : Collect admissible source evidence for this critical line before BoM can pass. power_distribution_dc_distribution_wet_mate_connector_set: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_thruster_array_thruster_set: Collect admissible source evidence for this critical line before BoM can pass. control_compute_communication_mission_computer_stack_mission_computer: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.8889	12 procurement-blocking closure row(s); 12 active sourcing row(s), 96 deferred row(s).	bom-close:actuation_kinematics_thruster_array_thruster_set: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication_acoustic_communications_acoustic_modem: Collect admissible source evidence for this critical line before	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				BoM can pass. bom-close:control_compute_communication_mission_computer_stack_mission_computer: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_storage_source_subsea_battery_pack_battery_pack: Collect admissible source evidence for this critical line before BoM can pass. bom-close:power_distribution_dc_distribution_wet_mate_connector_set: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/12; BoM total 0 GBP.	structure_containment_pressure_hull_s hell_pressure_hull: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_subsea_battery_pack_battery_pack : Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. power_distribution_dc_distribution_wet_mate_connector_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				BoM can pass. actuation_kinematic s_thruster_array_thr uster_set: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. control_compute_c ommunication_missi on_computer_stack _mission_computer: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 70 more	
prototype procurement policy	review	0	candidate_bom_revi ew_required; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Pressure hull	1 candidate	Pressure hull; Pressure hull manufacturer part number; Pressure hull catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Battery pack	1 candidate	Battery pack; Battery pack manufacturer part number; Battery pack catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Wet-mate connector set	1 candidate	Wet-mate connector set; Wet-mate connector set manufacturer part number; Wet-mate connector set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Thruster set	6 each	Thruster set; Thruster set manufacturer part number; Thruster set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Mission computer	1 candidate	Mission computer; Mission computer manufacturer part number; Mission computer catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Acoustic modem	1 candidate	Acoustic modem; Acoustic modem manufacturer part number; Acoustic modem catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
DVL	1 candidate	DVL; DVL manufacturer part number; DVL catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Inertial navigation unit	1 candidate	Inertial navigation unit; Inertial navigation unit manufacturer part number; Inertial navigation unit catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Depth and pressure sensor	1 candidate	Depth and pressure sensor; Depth and pressure sensor manufacturer part number; Depth and pressure sensor catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Forward sonar payload	1 candidate	Forward sonar payload; Forward sonar payload manufacturer part number; Forward sonar payload catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Leak detection sensor	1 candidate	Leak detection sensor; Leak detection sensor manufacturer part number; Leak detection sensor catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Recovery beacon	1 candidate	Recovery beacon; Recovery beacon manufacturer part number; Recovery beacon catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
PRESSURE-BOUNDARY-REVIEW	Depth-rated pressure boundary review	Project engineering gate	Depth rating calculation, hull material allowables, endcap seal design and proof-pressure plan.
SUBSEA-LEAK-ABORT-REVIEW	Leak detection and abort-state review	Project engineering gate	Leak sensor placement, abort state machine, battery isolation and recovery procedure.
EMC-MARINE-COMMS-REVIEW	Marine communications and EMC review	Project engineering gate	Acoustic modem integration, antenna/transducer mounting, cable shielding and data-loss behaviour.

Hazard	S	O	D	RPN	Mitigation
Pressure hull or seal failure at depth	10	3	4	120	Depth-rated pressure calculation, hydrostatic proof test, leak test and seal inspection workflow.
Loss of navigation or communications underwater	8	4	5	160	DVL/INS/depth fusion, acoustic link health monitoring and timed abort behaviour.
Unable to recover vehicle after abort	8	3	4	96	Recovery beacon, drop-weight/recovery procedure, deck handling points and mission reserve energy.

Section Issues And Score

bom

- blocker / critical_part_unpriced: Pressure hull is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Battery pack is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Wet-mate connector set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Thruster set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Mission computer is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Acoustic modem is critical but has no credible unit cost.

- blocker / critical_part_unpriced: DVL is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Inertial navigation unit is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Depth and pressure sensor is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Forward sonar payload is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Leak detection sensor is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Recovery beacon is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
							is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						